



Owen Bates

Computer Engineer

Westfield, NY
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Education

Rochester Institute of Technology

Bachelor of Science Degree - Computer Engineering – 5th Year - GPA: 3.8 (Graduating December 2025)

Skills

Technical Abilities/Software: Multimeter, Oscilloscope, Waveform Generator, ROS2, NAV2, Linux, Vivado, LTSPICE, Microsoft Office

Programming Languages: C, C++, GIT, Python, Java, VHDL ARM Assembly, MIPS Assembly, Verilog, MAKE, CMAKE, MATLAB, VBA, ABB Robotics, GML Language

Work Experience

- **Computer Engineering Intern – Cornell University CAIR Lab (June 2025 – September 2025)**
 - Integrated the Amiga robotic platform into the planting of a 4-acre vineyard, developing a custom waypoint program to enable autonomous row navigation.
 - Designed and implemented a pure-pursuit navigation system in ROS2, improving path-tracking accuracy and field performance.
 - Developed a variable-rate fertilizer control system, including a PyQt-based interface and custom PCB design, enabling automated nutrient application tailored to vineyard zones.
- **Computer Engineering Intern – Cornell University CAIR Lab (May 2024 – August 2024)**
 - Designed a navigation stack for an agricultural robot, Thorvald, to autonomously navigate complex agricultural landscapes.
 - Leveraged Python, C++, ROS2, and NAV2 to implement path planning, obstacle avoidance, and localization for reliable autonomous navigation.
- **Computer Engineering Intern – Cornell University CAIR Lab (June 2023 – December 2023)**
 - Upgraded the control system for an agricultural robot, improving responsiveness and reliability.
 - Utilized Python, C++, and ROS2, enabling advanced manoeuvrability in agricultural environments.
 - Developed a ROS2-based swarm communication system, enabling seamless interaction and coordination between multiple agricultural robots.

Projects (Check out My Website: <https://okb45.github.io/resumewebsite>)

- **Variable Rate Fertilizer Controller**
 - Designed and implemented a retrofit controller for a commercial fertilizer spreader, enabling variable-rate application based on real-time GNSS and zone mapping.
 - Developed control software in C for Raspberry Pi, integrating SPI-based actuator control, GNSS positioning, and shapefile-based field management.
 - Created a PyQt6 desktop application for system setup and operation and designed a custom PCB to consolidate wiring and hardware interfaces.
- **Hybrid Game Boy Advance– Teensy 4.1 & Zybo FPGA (Ongoing)**
 - Implementing a custom GBA emulator where game logic runs on a Teensy 4.1 microcontroller and graphics rendering is offloaded to a Zybo FPGA.
 - Designed a 16-bit parallel interface with finite state machine logic in VHDL to initialize and drive an ILI9486 display controller.
- **Multi-disciplinary Senior Design - (January 2025 – Present)**
 - Developing a sensor array and software for obstacle avoidance for transportation of Visually Impaired
 - Gaining experience in cross-functional teamwork, technical communication, and open-source development.